Mean Exit Times – Handout

Chapter 12 of Higham & Kloeden, *An Introduction to the Numerical Simulation of SDEs*

1 Motivation & Problem Statement

$$\mathrm{d}X(t) = f(X(t))\,\mathrm{d}t + g(X(t))\,\mathrm{d}W(t), \quad X(0) = X_0, \tag{12.1}$$

$$T_{\mathrm{a}} \coloneqq \inf\{t \;:\; X(t) = a\}, \qquad T_{\mathrm{b}} \coloneqq \inf\{t \;:\; X(t) = b\} \qquad [2, §1.2]$$

$$T_{\mathrm{exit}} \coloneqq T_{\mathrm{a}} \wedge T_{\mathrm{b}}, \qquad [2, \text{Lemma } 2.9]$$

$$T_{\mathrm{exit}}^{\mathrm{mean}} \coloneqq \mathbb{E}[T_{\mathrm{exit}}]$$

Goal: compute $T_{\mathrm{exit}}^{\mathrm{mean}}$ by simulation — and find out how accurate that is.

2 Monte Carlo – Euler-Maruyama

```
choose a stepsize $\Delta t$
choose a number of paths $M$
for $s = 1$ to $M$
  set $t_n = 0$ and $X_n = X_0$
  while $a < X_n < b$
    compute a $N(0,1)$ sample $\xi_n$
    replace $X_n$ by $X_n + \Delta t\, f(X_n) + \sqrt{\Delta t}\,\xi_n\, g(X_n)$
    replace $t_n$ by $t_n + \Delta t$
  end
  set $T^s_{\mathrm{exit}} = t_n - \frac{1}{2}\Delta t$        (midpoint of the last step)
end
set $a_M = \frac{1}{M}\sum_s T^s_{\mathrm{exit}}$
set $b_M^2 = \frac{1}{M-1}\sum_s \left(T^s_{\mathrm{exit}} - a_M\right)^2$
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$$95\,\%\ \text{CI:} \quad \left[a_M - 1.96\frac{b_M}{\sqrt{M}},\;\; a_M + 1.96\frac{b_M}{\sqrt{M}} \right] \qquad [1, §2.3]$$

Three sources of error.

(E1) sampling: sample mean $\neq$ expectation [1, §2.3]

(E2) discretization: EM path $\neq$ SDE path [1, §8–10]

(E3) missed exits: we check discretely at grid times $\{t_i\}$ **new!** [1, Fig. 12.1]
within (t_i, t_{i+1}) the path may leave (a, b) and return *unnoticed*

3 Analytical Baseline: from SDE to ODE

$$u(x) \coloneqq \mathbb{E}[T_{\mathrm{exit}} \mid X(0) = x]$$

$$P_x[\,\cdot\,] \coloneqq P[\,\cdot \mid X(0) = x\,], \qquad \mathbb{E}_x[\,\cdot\,] \coloneqq \mathbb{E}[\,\cdot \mid X(0) = x\,]$$

For a small Δt, the strong Markov property yields the renewal relation; rearranging this equation we find the average rate of change gives:

$$\begin{aligned} u(x) &= \Delta t + \mathbb{E}_x[\,u(X(\Delta t))\,] + o(\Delta t) \\ \frac{\mathbb{E}_x[\,u(X(\Delta t))\,] - u(x)}{\Delta t} &= -1 - \frac{o(\Delta t)}{\Delta t} \end{aligned}$$

Taking the continuous-time limit as $\Delta t \to 0$ defines the **infinitesimal generator $\mathcal{A}$**, which represents the expected rate of change of our function.

$$\mathcal{A}u(x) \coloneqq \lim_{\Delta t \downarrow 0} \frac{\mathbb{E}_x[\,u(X(\Delta t))\,] - u(x)}{\Delta t}$$

To compute $\mathcal{A}$ explicitly, apply Itô's lemma to the SDE (12.1).

$$\mathrm{d}u(X) = u'(X(t))\,\mathrm{d}X + \tfrac{1}{2}u''(X)\,(\mathrm{d}X)^2, \qquad [1, §6.2]$$

$$(\mathrm{d}W)^2 = \mathrm{d}t, \qquad (\mathrm{d}t)^2 = \mathrm{d}t \cdot \mathrm{d}W = 0.$$

$$u(X(\Delta t)) - u(x) = \int_0^{\Delta t} \Big(f u' + \tfrac{1}{2} g^2 u'' \Big)(X_s)\,\mathrm{d}s + \underbrace{\int_0^{\Delta t} (g\,u')(X_s)\,\mathrm{d}W_s}_{\text{mean-zero martingale}}\,.$$

For small Δt, the integral is dominated by the value of the integrand at the starting point $s = 0$ (where $X_0 = x$), multiplied by the width of the time step Δt:

$$\begin{aligned} \mathbb{E}_x[\,u(X(\Delta t))\,] - u(x) &= \mathbb{E}_x\left[\int_0^{\Delta t} \big(\tfrac{1}{2}g^2 u'' + f u'\big)(X_s)\,\mathrm{d}s \right]. \\ &= \big(\tfrac{1}{2}g(x)^2 u''(x) + f(x)\,u'(x)\big)\,\Delta t + o(\Delta t). \end{aligned}$$

Divide by Δt and let $\Delta t \downarrow 0$, by the definition of $\mathcal{A}$;
with $\mathcal{A}u = -1$ and $u = 0$ at the absorbing boundaries, we arrive at the ODE:

$$\begin{aligned} \mathcal{A}u(x) = \tfrac{1}{2}g(x)^2 u''(x) + f(x)u'(x) &= -1 \quad \text{on } (a, b), \\ u(a) = u(b) &= 0. \end{aligned} \tag{12.2}$$

4 The Toy Problem

$$\begin{aligned} \mathrm{d}X(t) &= \mu X(t)\,\mathrm{d}t + \sigma X(t)\,\mathrm{d}W(t), \\ X(t) &= X_0\, e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W(t)}, \\ \mathbb{E}[X(t)] &= X_0\, e^{\mu t}. \end{aligned}$$

[1, §5.2]

4.1 $u(x)$ for the toy problem

With $f(x) = \mu x$, $g(x) = \sigma x$, (12.2) becomes the Euler-type ODE

$$\tfrac{1}{2}\sigma^2 x^2\, u'' + \mu x\, u' = -1 \quad \text{on } (a,b), \qquad u(a) = u(b) = 0. \tag{12.3}$$

$$u(x) = \frac{1}{\frac{1}{2}\sigma^2 - \mu}\left(\log\frac{x}{a} - \frac{1-(x/a)^{1-2\mu/\sigma^2}}{1-(b/a)^{1-2\mu/\sigma^2}}\log\frac{b}{a}\right) \tag{12.4}$$

4.2 MC with exact path updates

We know the exact solution (section 4), so *line 7* of the pseudocode in section 2 can be improved — discretization error (E2) is switched off, isolating sampling (E1) + missed exits (E3):

instead of (EM step, section 2, line 7)

$$X_n \;\leftarrow\; X_n + \Delta t\, \mu X_n + \sqrt{\Delta t}\, \xi_n\, \sigma X_n$$

we use (exact update, from section 4)

$$X_n \;\leftarrow\; X_n \exp\big((\mu - \tfrac{1}{2}\sigma^2)\Delta t + \sqrt{\Delta t}\, \xi_n\, \sigma\big)$$

$$\mathrm{err}_{\Delta t} \coloneqq \big|a_M - T_{\mathrm{exit}}^{\mathrm{mean}}\big| \approx C\,\Delta t^{\,q}, \tag{12.5}$$

5 Outro & To Ponder

Some natural next questions to take with you

- What if instead of $T_{\mathrm{exit}}^{\mathrm{mean}}$, we wanted to know $\mathbb{E}_x[T_{\mathrm{a}}]$?
- What about the *probability that the process reaches a before b*, i.e. for $x \in (a,b)$, $P_x(T_{\mathrm{a}} < T_{\mathrm{b}}) = ?$
- $O(\Delta t^{1/2})$ is *very bad* — how to improve it?
- A *rigorous proof* of the order $O(\Delta t^{1/2})$??
- *Other methods* for mean exit times?
  - random-walk-based methods [4, §6]
  - multilevel Monte Carlo [1, §15]
  - a numerical method for the deterministic ODE (12.2)

References

[1] D. J. Higham and P. E. Kloeden, *An Introduction to the Numerical Simulation of Stochastic Differential Equations*, SIAM, Philadelphia, 2021.

[2] I. Karatzas and S. E. Shreve, *Brownian Motion and Stochastic Calculus*, Springer, New York, 1991.

[3] F. C. Klebaner, *Introduction to Stochastic Calculus with Applications*, Imperial College Press, London, 1998.

[4] G. N. Milstein and M. V. Tretyakov, *Stochastic Numerics for Mathematical Physics*, Springer, Berlin, 2004.

[5] R. Mannella, *A gentle introduction to the integration of stochastic differential equations*, in Stochastic Processes in Physics, Chemistry, and Biology, J. A. Freund and T. Poshel, eds., Springer, Berlin, 2000, pp. 353–364.

[6] K. M. Jansons and G. D. Lythe, *Exponential timestepping with boundary test for stochastic differential equations*, SIAM Journal on Scientific Computing, 24 (2003), pp. 1809–1822.

Check out the code

`https://github.com/cevheruysal/mean_exit_talk`

- `presentation.ipynb` (live figures, fresh samples on each run)
- `python scripts/make_figures.py` (all figures on demand).